ROLE 4 ACADEMIC REFERENCE COMPENDIUM • LITERATURE & TAXONOMY

LITERATURE REVIEW & EVIDENCE MATRIX: TRUST & REPUTATION IN BLOCKCHAIN CONSENSUS

Systematic Analysis of 12 Peer-Reviewed Consensus Architectures Across IEEE, ACM, Springer, Elsevier, and Wiley

Scope: 12 Verified Systems
Focus: Trust & Validator Selection
Standard: IEEE Citation Style

Analysis Depth: 10 Academic Dimensions
Taxonomy: PoS, DPoS, BFT, DAG
Integration: Supervised Machine Learning

Date: September 2026
Lead Author: Role 4 (Research Lead)
Target: Academic Thesis & Defense

Document Purpose & Academic Scope

This document serves as the authoritative academic evidence compendium for the project. It provides an exhaustive, multi-dimensional profile of twelve (12) seminal and cutting-edge peer-reviewed research papers. Each entry details the core problem addressed, underlying consensus mechanics, mathematical reputation formulations, behavioral telemetry inputs, validator selection impacts, architectural trade-offs, machine learning utilization, and exact relevance to our experimental pipeline.

1. Comprehensive Comparative Literature Matrix

Paper / System	Year	Base Consensus	Trust / Reputation Formulation	Behavioral Telemetry Signals	ML?	Key Advantage	Key Limitation
PoIR [Windiatmaja et al.]	2023	PoS / PBFT	Bi-LSTM temporal trust regression	Packet delivery ratio, delay, node drop rate	YES	Captures non-linear temporal behavioral trends	High inference latency; ignores economic stake
DPoR [Do et al.]	2019	DPoS	Reputation replacing coin voting	Block generation rate, valid txs, uptime	NO	Mitigates plutocratic coin-holding cartels	Susceptible to strategic whitewashing attacks
FPoR [Zhang & Huang]	2022	PoR	Credit scoring with aging decay	Verification accuracy, consensus delay	NO	Aging prevents permanent validator cartels	Linear formulas cannot detect camouflage
RL Validator [Harris]	2026	PoS	Deep RL policy scheduling	Uptime ratio, network latency, slot miss rate	YES	Adapts dynamically to network jitter & stragglers	Optimizes liveness; blind to Byzantine fraud
PPoR-X [Bou Abdo et al.]	2020	PoW/PoS	Multi-attribute dynamic trust score	Activity rate, validation correctness, latency	NO	Rigorous formal penalties for execution failure	Extreme cold-start penalty for newcomers
DC-DPoS [Zhao & Dai]	2023	DPoS	Sliding-window credit rating	Processing speed, dispute rates, uptime	NO	Rapid isolation of transient hardware faults	Vulnerable to cyclical on-off gaming attacks
GT-LWC [Xi et al.]	2025	PBFT	Direct trust + recommendation matrix	Proposal validity, message delays, peer votes	NO	Robust against isolated Byzantine proposals	$O(N^2)$ message complexity; collusion cliques
Sliding FBA [Nugroho et al.]	2026	Fed. BFT	Temporal sliding window evaluation	Quorum participation, vote divergence, ping	NO	Filters unresponsive nodes from quorum slices	Short memory windows reset malicious histories
IoV Consensus [Kang et al.]	2019	DPoS	Subjective logic + contract theory	Data verification accuracy, delay, frequency	NO	Incentivizes high-quality edge sensing	Excessive contract negotiation overhead
MeDiStore [Sanober & Anwar]	2025	PoS	Storage integrity & attestation weight	Proof of storage, response time, attestation	NO	Directly binds data availability to consensus	Domain-specific to healthcare storage systems
DAG-PoR [Dong et al.]	2024	DAG	Graph attestation credibility metric	Cross-validation match, propagation speed	NO	High throughput in asynchronous networks	Graph traversal overhead; complex ordering

EigenTrust [Kamvar et al.]	2003	P2P Net	Normalized eigenvector convergence	Satisfactory vs. unsatisfactory downloads	NO	Foundational math for distributed reputation	Zero-stake Sybil vulnerability in open nets

2. Detailed System Profiles (Systems 1 to 6)

System 1: PoIR (Windiatmaja, Hanggoro, & Salman, 2023)

- **Full Citation:** J. Windiatmaja, R. Hanggoro, and M. Salman, "PoIR: A Node Selection Mechanism in Reputation-Based Blockchain Consensus Using Bidirectional LSTM Regression Model," *Computers, Materials & Continua*, vol. 76, no. 1, pp. 411-430, 2023. DOI: `10.32604/cmc.2023.041152`.
- **Problem Solved:** Eliminating sluggish, unreliable edge nodes from IoT blockchain consensus committees to minimize latency and packet drops.
- **Base Consensus:** Modifies hybrid PoS / Practical Byzantine Fault Tolerance (PBFT).
- **Trust Calculation Method:** Telemetry windows are processed by a Bidirectional LSTM neural network that predicts next-epoch continuous reputation scores.
- **Behavioral Signals Used:** Packet delivery ratio, average transmission delay, node drop frequency, and CPU load.
- **Validator Selection Impact:** The top-K highest-reputation nodes are deterministically elected to the active validation committee.
- **Advantages:** Captures complex, non-linear temporal correlations and trend reversals in node performance.
- **Weaknesses / Limitations:** High neural network training and inference overhead; selecting solely on reputation creates severe Sybil vulnerability because no capital stake is required.
- **ML Component:** Bi-LSTM Recurrent Neural Network.
- **Relevance to Our Project:** Validates using machine learning to predict validator trustworthiness from operational telemetry; provides empirical grounding for using response time and drop rates as key signals.

System 2: Delegated Proof of Reputation (Do, Nguyen, & Pham, 2019)

- **Full Citation:** T. Do, T. Nguyen, and H. Pham, "Delegated Proof of Reputation: a novel Blockchain consensus," in *Proc. 2019 Int. Electronics Communication Conf. (IECC)*, ACM, 2019, pp. 68-73. DOI: `10.1145/3343147.3343160`.
- **Problem Solved:** Breaking the plutocratic concentration of consensus power in DPoS where wealthy coin holders control witness elections.
- **Base Consensus:** Delegated Proof of Stake (DPoS).
- **Trust Calculation Method:** A linear heuristic combining cumulative blocks produced, valid transactions confirmed, and uptime ratio.
- **Behavioral Signals Used:** Block generation success rate, uptime percentage, and verified transaction throughput.
- **Validator Selection Impact:** Nodes are ranked strictly by reputation index; top-ranked nodes become the block producing delegates.
- **Advantages:** Decouples consensus participation from passive token accumulation, incentivizing active operational maintenance.
- **Weaknesses / Limitations:** Highly susceptible to identity whitewashing (burning low reputation to restart) and lacks machine learning to detect coordinated attacks.
- **ML Component:** None (Rule-based weighted formula).
- **Relevance to Our Project:** Directly informs our selection of uptime and verification correctness as primary behavioral indicators.

System 3: Fair Proof-of-Reputation (Zhang & Huang, 2022)

- **Full Citation:** L. Zhang and Y. Huang, "FPoR: Fair proof-of-reputation consensus for blockchain," *ICT Express*, Elsevier, vol. 8, no. 4, pp. 581-586, 2022. DOI: `10.1016/j.icte.2022.11.007`.
- **Problem Solved:** The "rich-get-richer" reputation monopoly where early honest nodes become unchallengeable consensus leaders.
- **Base Consensus:** Standalone Proof of Reputation (PoR).
- **Trust Calculation Method:** Credit scoring incorporates an exponential temporal decay factor: $R(t) = R(t-1) \cdot e^{-\lambda \Delta t} + \Delta R$.
- **Behavioral Signals Used:** Verification accuracy, consensus response latency, and historical tenure.
- **Validator Selection Impact:** Proposer selection is conducted via a pseudo-random lottery weighted by decayed reputation.
- **Advantages:** Temporal decay forces incumbent validators to continually deliver high performance or lose selection priority.
- **Weaknesses / Limitations:** Decay parameters require fine-tuning; vulnerable to cyclical attackers who ramp up performance immediately prior to major exploits.
- **ML Component:** None (Exponential decay equations).
- **Relevance to Our Project:** Emphasizes the need to prevent early high-trust validators from dominating selection indefinitely.

System 4: Availability-Aware RL Validator Scheduling (Harris, 2026)

- **Full Citation:** J. Harris, "Availability-Aware Reinforcement Learning for Validator Scheduling in Proof-of-Stake Blockchains," in *2026 IEEE Int. Conf. Blockchain and Cryptocurrency (ICBC)*, 2026. DOI: `10.1109/icbc67748.2026.11575544`.
- **Problem Solved:** Orphaned blocks and network stalls caused by scheduling validators during unexpected offline periods or high network jitter.
- **Base Consensus:** Proof of Stake (PoS).
- **Trust Calculation Method:** Deep Q-Network (DQN) / Proximal Policy Optimization (PPO) agent observes historical validator state vectors and schedules slots to maximize availability.
- **Behavioral Signals Used:** Rolling uptime, network ping latency, slot proposal miss rate, and attestation arrival delta.
- **Validator Selection Impact:** Dynamically adjusts slot assignment probabilities, skipping nodes currently experiencing network instability.
- **Advantages:** Reduces missed block slots by over 40% in empirical testnets under fluctuating ISP conditions.
- **Weaknesses / Limitations:** Focuses exclusively on availability and liveness; provides zero defense against malicious Byzantine nodes validating fraudulent transactions.
- **ML Component:** Deep Reinforcement Learning (Actor-Critic / PPO).
- **Relevance to Our Project:** Demonstrates that latency and uptime are strong predictive signals; highlights that availability must be paired with verification correctness to detect Byzantine nodes.

System 5: Permissionless Proof-of-Reputation-X (Bou Abdo et al., 2020)

- **Full Citation:** J. Bou Abdo, R. El Sibai, and K. Demerjian, "Permissionless proof-of-reputation-X: A hybrid reputation-based consensus algorithm for permissionless blockchains," *Transactions on Emerging Telecommunications Technologies*, Wiley, vol. 31, no. 12, 2020. DOI: `10.1002/ett.4148`.
- **Problem Solved:** Extending reputation models into open, permissionless blockchains while preventing Sybil identities from overtaking consensus.

- **Base Consensus:** Hybrid Proof of Work / Proof of Stake.
- **Trust Calculation Method:** Comprehensive score aggregating historical activity volume, block verification accuracy, and propagation latency.
- **Behavioral Signals Used:** Node active tenure, block propagation delay, transaction verification correctness, and attestation completeness.
- **Validator Selection Impact:** Only nodes exceeding a minimum reputation threshold are admitted to the block proposal lottery.
- **Advantages:** Rigorous mathematical formulation of execution failure penalties; effectively filters persistent low-quality nodes.
- **Weaknesses / Limitations:** New honest nodes face severe cold-start barriers; high barrier to entry can decrease validator decentralization.
- **ML Component:** None (Statistical aggregation functions).
- **Relevance to Our Project:** Directly informs our verification correctness feature and emphasizes the critical importance of a fair cold-start strategy ($T_0 = 0.5$).

System 6: Dynamic Credit DPoS (Zhao & Dai, 2023)

- **Full Citation:** M. Zhao and C. Dai, "Safe and Efficient Delegated Proof of Stake Consensus Mechanism Based on Dynamic Credit in Electronic Transaction," *Journal of Internet Technology*, vol. 24, no. 1, pp. 101-112, 2023. DOI: `10.53106/160792642023012401012`.
- **Problem Solved:** Mitigating witness node collusion and sluggish transaction confirmation in high-throughput electronic transaction ledgers.
- **Base Consensus:** Delegated Proof of Stake (DPoS).
- **Trust Calculation Method:** Sliding temporal window computing credit scores from transaction confirmation throughput, dispute frequency, and uptime.
- **Behavioral Signals Used:** Transaction processing throughput, dispute/reversal rates, node heartbeat responses.
- **Validator Selection Impact:** Witness voting power is dynamically scaled by credit scores; low-credit witnesses are automatically disqualified.
- **Advantages:** Rapid detection and isolation of newly misbehaving or compromised witnesses.
- **Weaknesses / Limitations:** Heuristic thresholds can be gamed by cyclical adversaries who behave honestly just long enough to maintain credit.
- **ML Component:** None (Sliding-window arithmetic).
- **Relevance to Our Project:** Justifies our use of rolling behavioral evaluation windows and monitoring transaction voting outcomes.

3. Detailed System Profiles (Systems 7 to 12)

System 7: Global Trust Lightweight Consensus (Xi, Xu, & Zou, 2025)

- **Full Citation:** Z. Xi, J. Xu, and Y. Zou, "A global trust-based blockchain lightweight consensus mechanism," *Blockchain: Research and Applications*, Elsevier, vol. 6, no. 2, 2025. DOI: `10.1016/j.bcra.2025.100322`.
- **Problem Solved:** Scalability bottlenecks and Byzantine leader attacks in PBFT-based consortium blockchains.
- **Base Consensus:** Practical Byzantine Fault Tolerance (PBFT).
- **Trust Calculation Method:** Direct interaction trust combined with EigenTrust-style indirect peer recommendation matrices.

- **Behavioral Signals Used:** Proposal block validity, message propagation delays, and peer signature validation consistency.
- **Validator Selection Impact:** Consensus committee leaders are selected exclusively from the top decile of global trust ranks.
- **Advantages:** Isolates Byzantine leaders before they can propose invalid blocks, eliminating costly PBFT view-change protocols.
- **Weaknesses / Limitations:** $O(N^2)$ message complexity for peer recommendation gossip; vulnerable to colluding peer cliques.
- **ML Component:** None (Matrix linear algebra).
- **Relevance to Our Project:** Confirms that proposal validity and message delay are foundational signals; illustrates why objective telemetry is preferable to subjective peer recommendation gossip.

System 8: Sliding Window FBA Evaluation (Nugroho et al., 2026)

- **Full Citation:** A. Nugroho, A. Isnanto, and R. BayuAji, "Sliding Window-Based Evaluation of Validator Nodes in Blockchain Federated Byzantine Agreement," in *2025 IEEE 9th Int. Conf. Software Engineering & Computer Systems (ICSECS)*, 2026. DOI: `10.1109/icsecs65227.2025.11279049`.
- **Problem Solved:** Deadlock in Federated Byzantine Agreement (FBA) networks (e.g. Stellar, Ripple) caused by unresponsive quorum slice members.
- **Base Consensus:** Federated Byzantine Agreement (FBA).
- **Trust Calculation Method:** Temporal sliding window computing reliability ratings based on quorum participation frequency and voting consistency.
- **Behavioral Signals Used:** Ping latency, quorum vote response latency, vote divergence from final consensus outcome.
- **Validator Selection Impact:** Nodes dynamically adjust their local quorum slices to remove nodes whose reliability rating drops below threshold.
- **Advantages:** Prevents network stalls without requiring centralized validator coordination.
- **Weaknesses / Limitations:** Short sliding windows can erase historical records of past Byzantine behavior.
- **ML Component:** None (Sliding moving average).
- **Relevance to Our Project:** Validates tracking voting divergence (transaction outcome approval/decline) as a core telemetry feature.

System 9: IoV Contract-Reputation Consensus (Kang et al., 2019)

- **Full Citation:** J. Kang, Z. Xiong, D. Niyato, D. Ye, D. I. Kim, and J. Shen, "Towards Secure Blockchain-enabled Internet of Vehicles: Optimizing Consensus Management Using Reputation and Contract Theory," *IEEE Transactions on Vehicular Technology*, vol. 68, no. 3, pp. 2906-2920, 2019. DOI: `10.1109/TVT.2019.2898705`.
- **Problem Solved:** Securing vehicular ad-hoc networks (VANETs) against malicious vehicles broadcasting false sensory telemetry.
- **Base Consensus:** Delegated Proof of Stake (DPoS).
- **Trust Calculation Method:** Subjective logic model evaluating multi-weight historical interactions combined with contract-theoretic economic incentives.
- **Behavioral Signals Used:** Data verification accuracy, interaction frequency, message delivery latency, vehicle mobility tenure.
- **Validator Selection Impact:** High-reputation vehicles are contracted as consensus standby miners and awarded dynamic block fees.
- **Advantages:** Mathematically sound economic equilibrium that optimizes validator effort and data honesty.

- **Weaknesses / Limitations:** High computational complexity for contract negotiation; unsuited for general-purpose decentralized ledgers.
- **ML Component:** None (Subjective Logic & Contract Theory).
- **Relevance to Our Project:** Reinforces the synergy between economic incentives (stake) and historical behavioral fidelity (reputation).

System 10: MeDiStore Reputation-Weighted PoS (Sanober & Anwar, 2025)

- **Full Citation:** S. Sanober and N. Anwar, "MeDiStore trust protocol integrating reputation based proof of stake consensus for enhanced security of blockchain networks in electronic health record management," *Discover Computing*, Springer, vol. 28, no. 1, 2025. DOI: `10.1007/s10791-025-09540-2`.
- **Problem Solved:** Ensuring storage integrity and access compliance in decentralized electronic health record (EHR) systems.
- **Base Consensus:** Proof of Stake (PoS).
- **Trust Calculation Method:** Combines cryptographic proof of data possession with validator attestation fidelity into a composite reputation factor.
- **Behavioral Signals Used:** Cryptographic storage proof validity, attestation response time, block verification accuracy.
- **Validator Selection Impact:** Proposer selection probability is scaled by combining locked token stake and historical storage reputation.
- **Advantages:** Directly ties consensus voting rights to active service delivery (storage proof) rather than passive coin ownership.
- **Weaknesses / Limitations:** Tailored specifically to storage blockchains; lacks machine learning pattern classification.
- **ML Component:** None (Multi-factor algebraic weighting).
- **Relevance to Our Project:** Closest conceptual analog in recent literature for multiplying economic stake by a behavioral reputation factor ($Stake imes Trust$).

System 11: Hierarchical DAG-PoR (Dong, Wu, & Li, 2024)

- **Full Citation:** Z. Dong, W. Wu, and X. Li, "Trustworthy VANET: Hierarchical DAG-Based Blockchain Solution with Proof of Reputation Consensus Algorithm," in *2023 IEEE Int. Conf. Blockchain*, 2024. DOI: `10.1109/blockchain60715.2023.00030`.
- **Problem Solved:** Linear blockchain throughput bottlenecks and selfish transaction referencing in Directed Acyclic Graph (DAG) protocols.
- **Base Consensus:** Directed Acyclic Graph (DAG) Consensus.
- **Trust Calculation Method:** Graph attestation credibility score based on peer validation cross-referencing within the DAG tangle.
- **Behavioral Signals Used:** Transaction cross-verification accuracy, tip attachment speed, node response latency.
- **Validator Selection Impact:** Tips with higher reputation weights receive priority referencing, guaranteeing rapid confirmation for honest transactions.
- **Advantages:** Massive parallel throughput without single-proposer bottlenecks.
- **Weaknesses / Limitations:** Significant graph traversal and tip selection computational overhead.
- **ML Component:** None (Graph topology metrics).
- **Relevance to Our Project:** Proves that verification correctness and latency are universally applicable across both linear chains and DAG architectures.

System 12: EigenTrust P2P Reputation (Kamvar, Schlosser, & Garcia-Molina, 2003)

- **Full Citation:** S. D. Kamvar, M. T. Schlosser, and H. Garcia-Molina, "The EigenTrust algorithm for reputation management in P2P networks," in *Proc. 12th Int. Conf. World Wide Web (WWW)*, 2003, pp. 640-651. DOI: `10.1145/775152.775242`.
- **Problem Solved:** Minimizing the spread of inauthentic or malicious files in peer-to-peer file sharing networks.
- **Base Consensus:** Peer-to-Peer Distributed Trust (Foundational baseline).
- **Trust Calculation Method:** Normalized local trust ratings converge to a global trust vector via stationary probability distribution of a Markov chain ($t^{(k+1)} = C^T t^{(k)}$).
- **Behavioral Signals Used:** Satisfactory vs. unsatisfactory file download transactions.
- **Validator Selection Impact:** Peers select download sources with probability proportional to their global trust values.
- **Advantages:** Mathematically elegant; provably minimizes impact of malicious peer groups who mutually praise one another.
- **Weaknesses / Limitations:** Lacks economic skin-in-the-game; vulnerable to large-scale Sybil swarms where dummy peers outnumber honest nodes.
- **ML Component:** None (Power iteration linear algebra).
- **Relevance to Our Project:** Foundational reference for distributed trust calculation; illustrates why economic stake ($Stake$) must anchor behavioral trust ($Trust$) to prevent zero-cost Sybil attacks.

4. Methodological Taxonomy Across Consensus Families

Consensus Family	Typical Systems	How Trust is Integrated	Primary Failure Mode
Proof-of-Stake (PoS)	Ethereum Gasper, MeDiStore, Harris (2026), Our System	Multiplies or filters stake-based lottery probabilities using behavioral metrics.	Wealthy malicious nodes purchase high proposal share if trust weighting is too weak.
Delegated Proof-of-Stake (DPoS)	DPoR, DC-DPoS, IoV Consensus, BitShares	Replaces token-weighted delegate voting with reputation scores, or filters active witnesses.	Witness collusion and strategic on-off voting cycles around credit thresholds.
Byzantine Fault Tolerance (BFT / PBFT)	GT-LWC, Sliding FBA, Tendermint	Leader election dynamically filters untrusted nodes; quorum slices drop slow peers.	High message complexity ($O(N^2)$); deadlock if quorum slices diverge excessively.
Directed Acyclic Graph (DAG)	Hierarchical DAG-PoR, IOTA Tangle	Transaction tips referenced according to cumulative author reputation.	Graph traversal complexity and asynchronous tip selection race conditions.

5. Research Gap Analysis & Team Innovation

The Fundamental Academic Void Addressed by Our Project

Existing literature is split into two opposing extremes:

1. **Pure Capital Models (Standard PoS):** Blindly assume capital equals honesty, ignoring key compromise and Byzantine unreliability.
2. **Pure Reputation Models (DPoR, PoR):** Eliminate stake entirely, leaving the protocol vulnerable to costless Sybil identity generation.

Systems that attempt to combine stake and reputation (e.g. MeDiStore) rely on static, linear arithmetic weighting that fails against adaptive Byzantine camouflage. Furthermore, almost no literature evaluates trust-aware consensus via a **strictly controlled, head-to-head comparison on the exact same underlying behavioral trajectory**.

Our Contribution: We implement a dual-track comparative simulation (100 validators, 1,000 transactions) that trains an interpretable **Random Forest model** on multidimensional behavioral telemetry ($uptime, latency, verification correctness, voting disposition, anomaly flags$) and couples it multiplicatively with capital collateral ($W_i = ext{Stake}_i imes ext{Trust}_i$). This provides definitive empirical evidence on whether behavioral ML can reduce malicious validator selection without sacrificing Sybil security.